THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

Recommendation Pitch

Data Analytics Career Accelerator

Team 11

Bev Grantham | Jessy Pardi | Emmanuel Wagaba | Davide Casarin

StudyGroup

Background

Purpose: Evaluate **discount effectiveness** and identify underutilised opportunities.

Key Focus: Optimise discount allocation to **boost enrollments** and **progression** rates

Scope: Analyse **historical data** across regions, programmes, and lead sources

Audience: Regional sales teams, programmes coordinators, and leadership

Business Outcome: Maximize **ROI** while supporting **student success** and growth.

Data analysis and visualisation approach

Step 1

Student Flow Analysis & Key Metrics

Tracked student progression across **key stages** (Offer → Confirmation → Arrival → Registration).

Identified major **drop-off points** and calculated conversion rates across programmes, regions, and lead sources.

Created **summary tables and visualisations** to highlight bottlenecks and optimize student retention strategies.

Step 2

Discount Effectiveness & Correlation Analysis

Explored **discount distribution across centers, agents, programmes, and regions** to determine impact on enrollments.

Measured correlation between **discounts, confirmations, and final registrations**, identifying **high- and low-impact discount strategies**.

Step 3

Predictive Modeling for Confirmation & Net Value

Developed **a Random Forest model** to predict **confirmation rates based on discount levels, lead source, and programme type**.

Simulated **how adjusting discount allocations** would impact student registrations and financial return.

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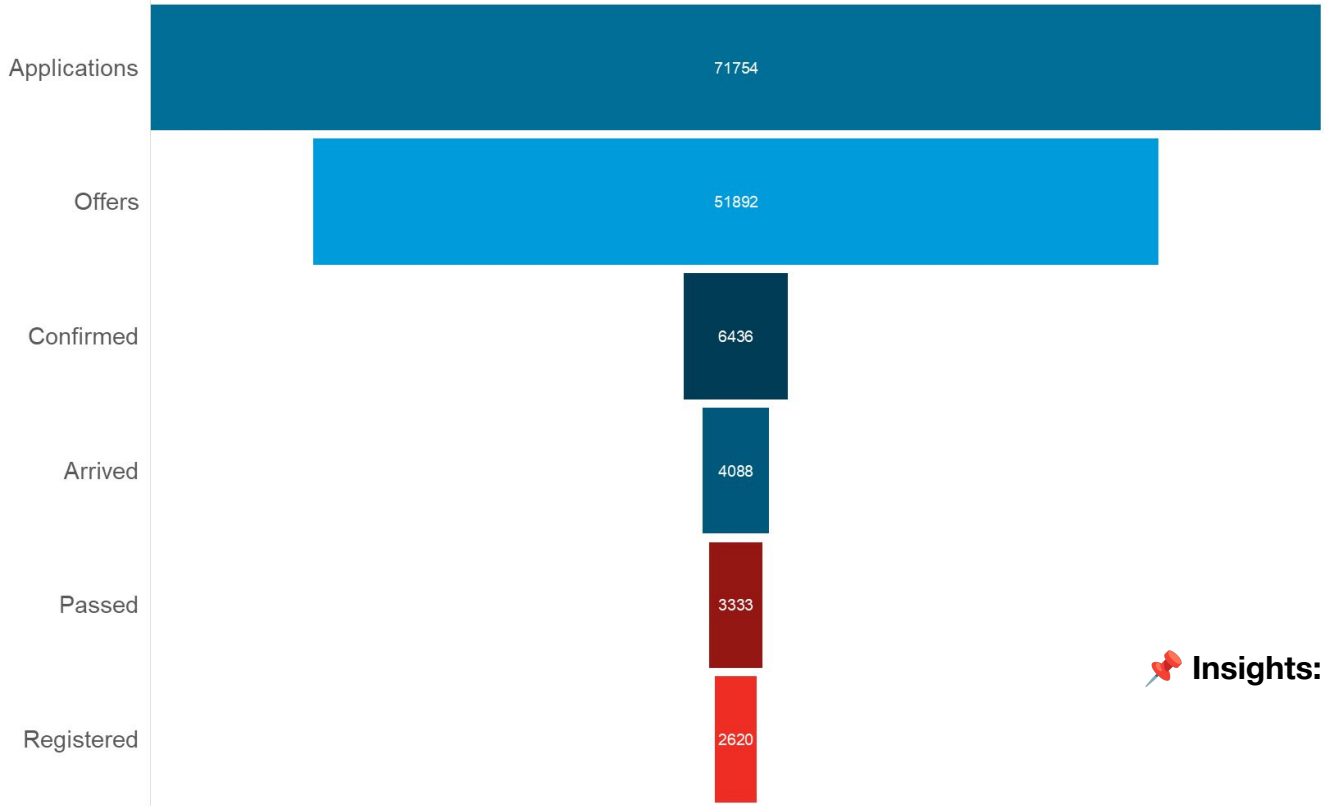
The Student Journey



71,754 applications were made for three pathway courses across 15 centres.

51,892 offers were made

6,436 offers were accepted and the student confirmed their place.

4,088 students arrived and **3,333 passed** the pathway course.

2,620 students went on to register with the University

📌 **Insights:** Consider recording cancellation reasons when students leave the pathway.

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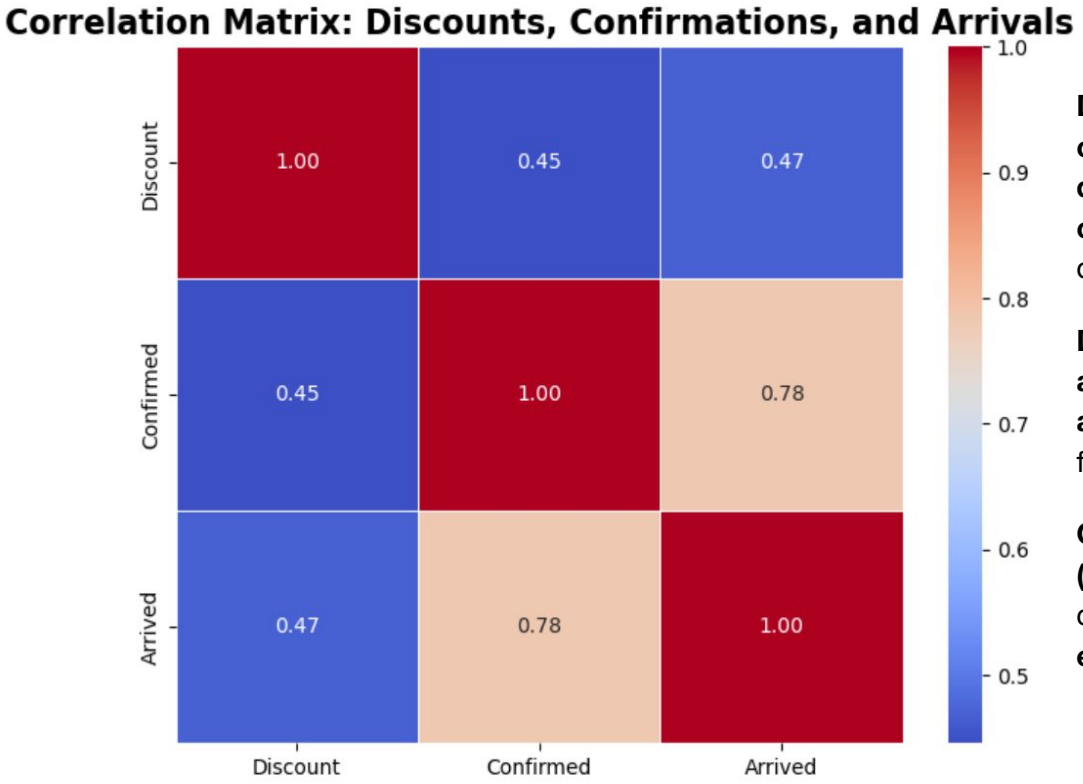
Correlation matrix

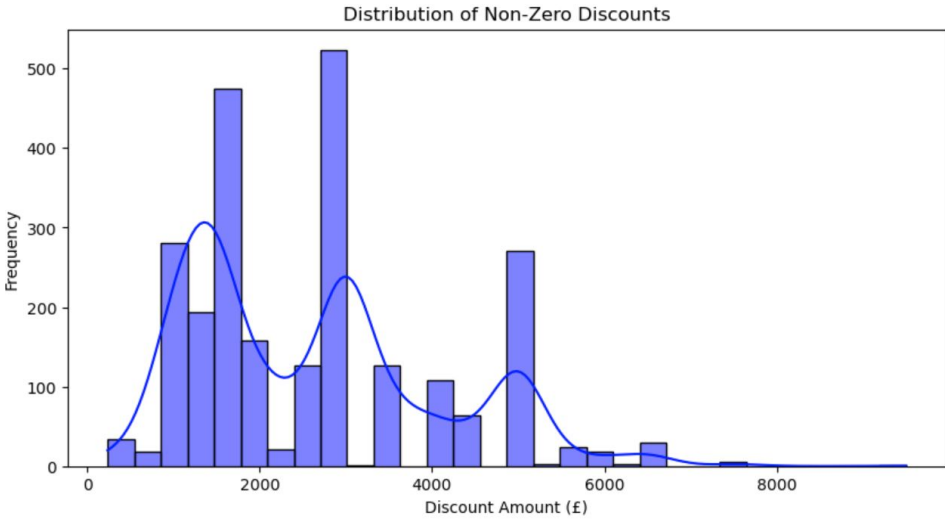


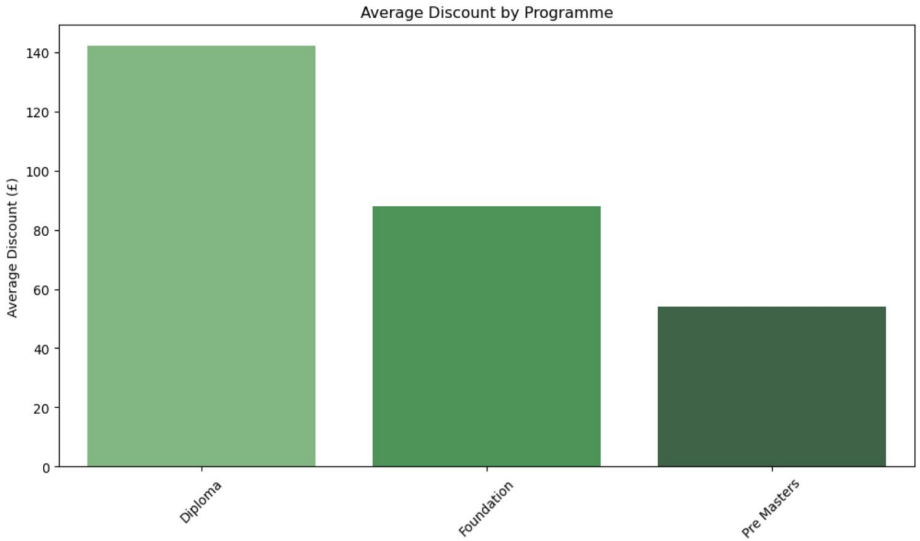
Discounts moderately correlate with confirmations (~0.45): This confirms that **higher discounts tend to increase student confirmations**, but other factors also influence decisions.

Discounts also moderately correlate with arrivals (~0.47) : Students who receive discounts **are slightly more likely to arrive**, meaning financial incentives play a role in final enrollment.

Confirmations strongly correlate with arrivals (~0.78): This means that most students who commit to a program **actually follow through and enroll**.

Discount Distribution Analysis





- Most discounts fall between **£500 and £4,000**.
- Discounts are allocated to **early-stage students primarily**, suggesting that **they require more financial incentives** to enroll.

📌 **Insights:**

- Focus discounts more on **Foundation & Diploma students** to boost early-stage enrollments.
- For Pre-Masters, shift to **non monetary incentives**

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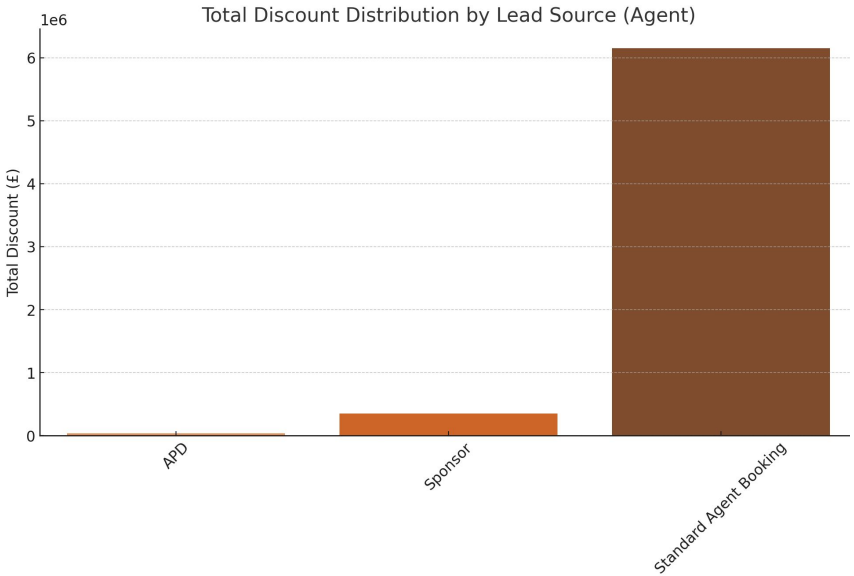
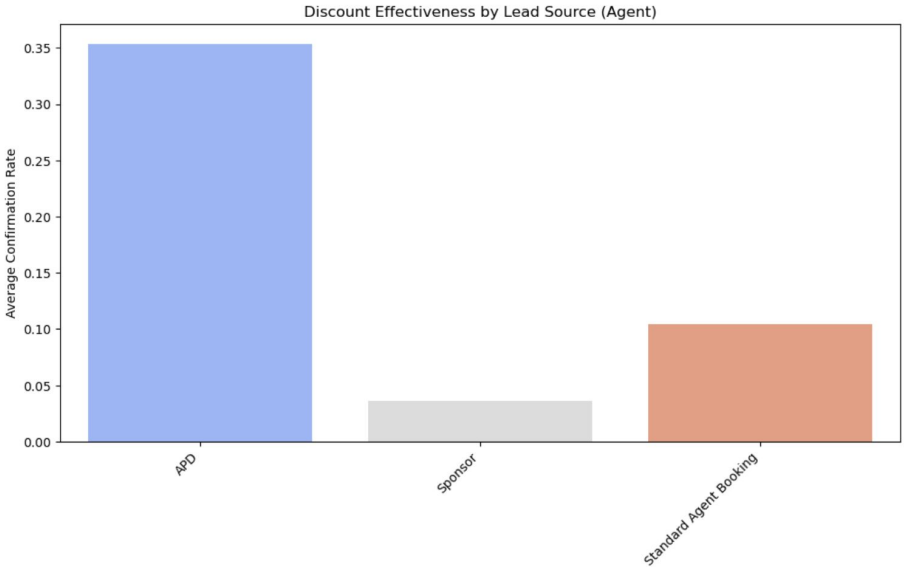
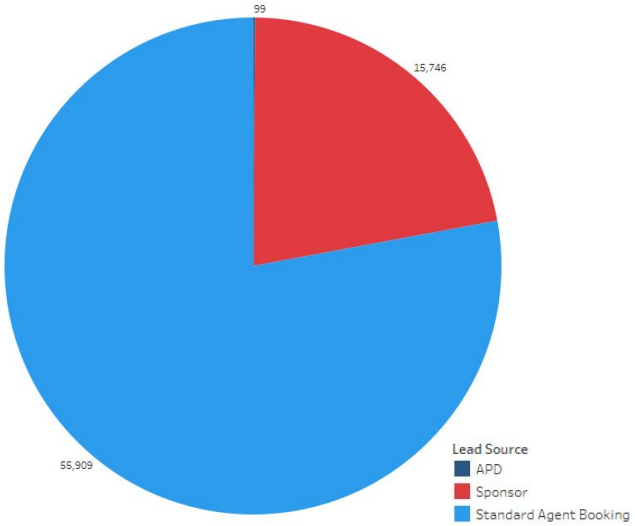
Discount Distribution & Effectiveness by Lead Source

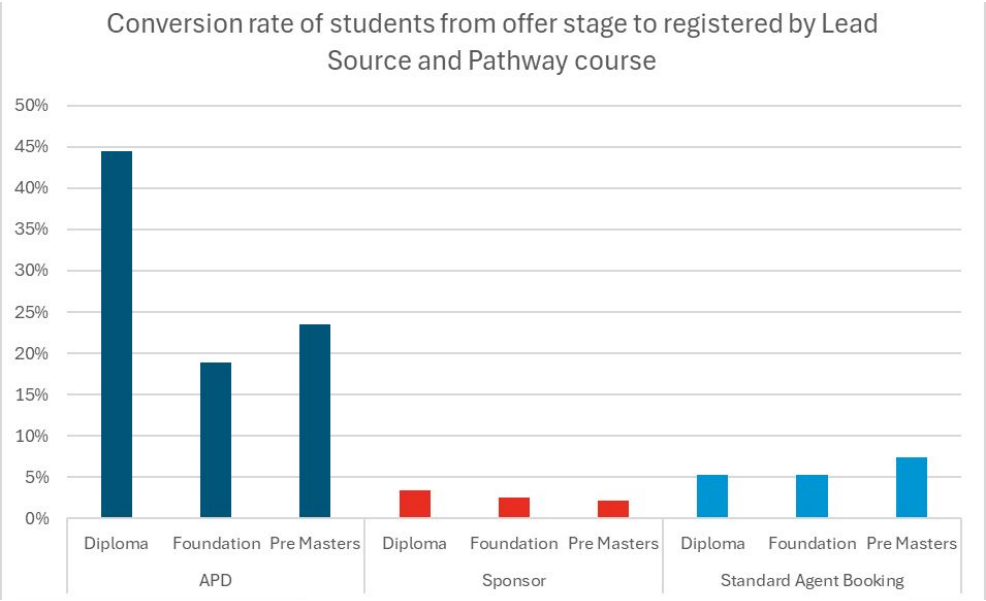




- **Standard Agent Bookings** receive the **highest total discounts**, yet their **confirmation rates are relatively low** = agent-led recruitment may not be the most **cost-effective**.
- **APD students** receive **fewer total discounts** but show **the highest confirmation rates**, suggesting stronger commitment even with lower financial incentives.
- Note: **APD students receive higher average discounts** = well-targeted financial aid leads to strong student commitment.

Conversion rates by Lead Source and Pathway course





📌 **Insights:**

- **Reduce discount reliance for Standard Agent Bookings** and explore performance-based incentives instead.
- **Increase APD enrollments**, as they convert better with lower discount spending.

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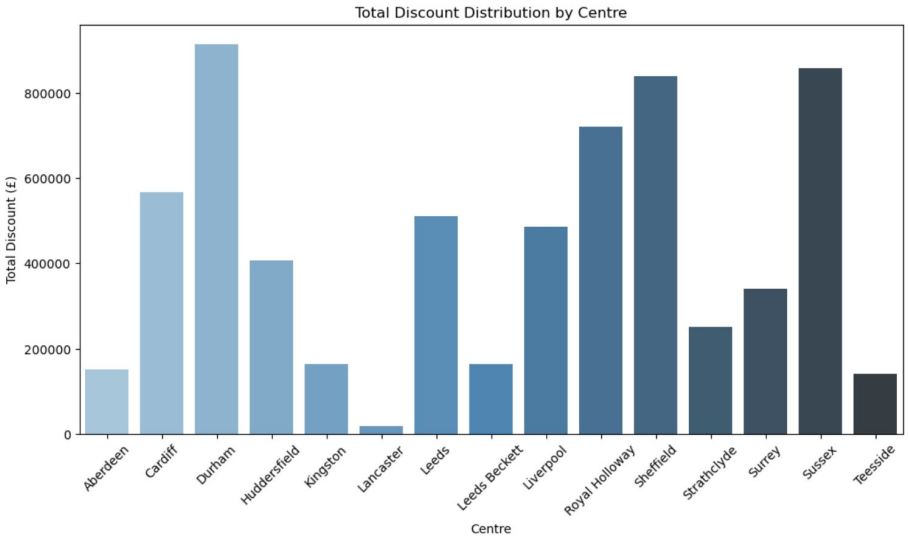
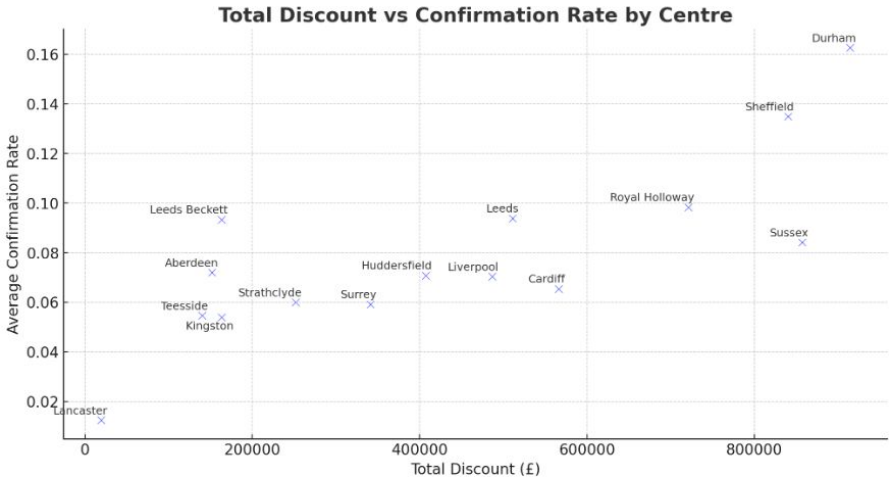
Discount Distribution & Effectiveness by Center





- Some centers (**Durham, Sheffield, Sussex**) receive significantly **higher total discounts**. **Discount allocation is uneven**, suggesting that some centers require **more financial incentives** than others.
- Centers with higher total discounts achieve **higher confirmation rates.**

📌 **Insights:**

- **Maintain or increase discounts** for high-performing centers where discounts **are proven to drive enrollment**.
- **Reallocate discounts for low-impact centers**, testing alternative incentives.

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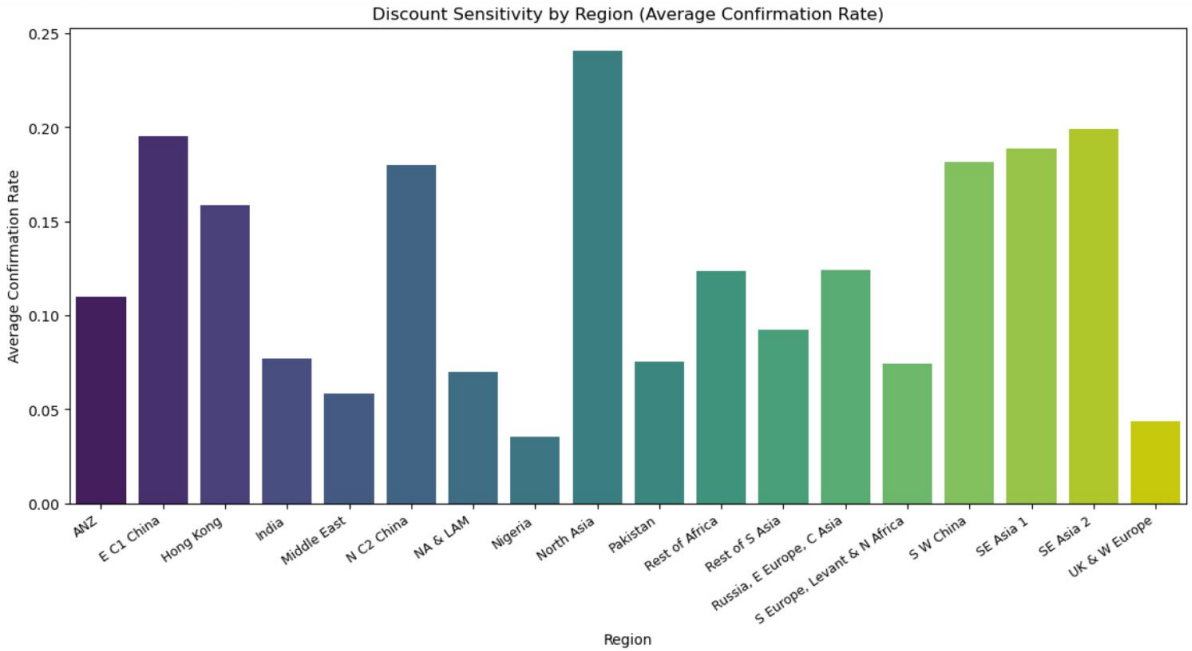
Discount Sensitivity by Region



- **Pakistan, Middle East, and the UK & Western Europe=** highest total discounts.
- **North Asia and Southeast Asia 2** = **highest average discounts,** suggesting that students from these regions may require higher incentives to enroll.

Looking at Confirmation rate:

- **North Asia and Southeast Asia** show high confirmation rates with discounts.
- Some regions (e.g., Nigeria, Middle East) have lower confirmation rates despite receiving discounts, suggesting that **discounts alone may not be enough** of a decision factor.

📌 **Insights:**

- **Increase discount allocations for high-response regions** (North Asia, Southeast Asia) where financial incentives **clearly drive enrollment**.

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Predicting Enrollment with Data-Driven Insights

Predict enrollment likelihood using a Random Forest Classifier segmented by areas

1

Preparation:

- Segmented Students into Areas for more accurate modelling

- Selection of relevant Features for the model

2

Trained & Evaluated a Random Forest Classifier per area

Net Value	Represents the net value of tuition after applying any discount, which directly affects the financial burden on students.
Parent Agent (Group URN)	The specific Parent Agent that sent the application (identified as Group URN in the dataset)
Discount Amount:	The absolute value of the discount provided, highlighting whether larger discounts lead to better enrollment rates.
Centre	Reflects the influence of the center where the student is enrolled. This accounts for regional and institutional preferences.
Nationality	Identifies how a student's nationality may affect enrollment likelihood, particularly for regions with varying scholarship sensitivities.

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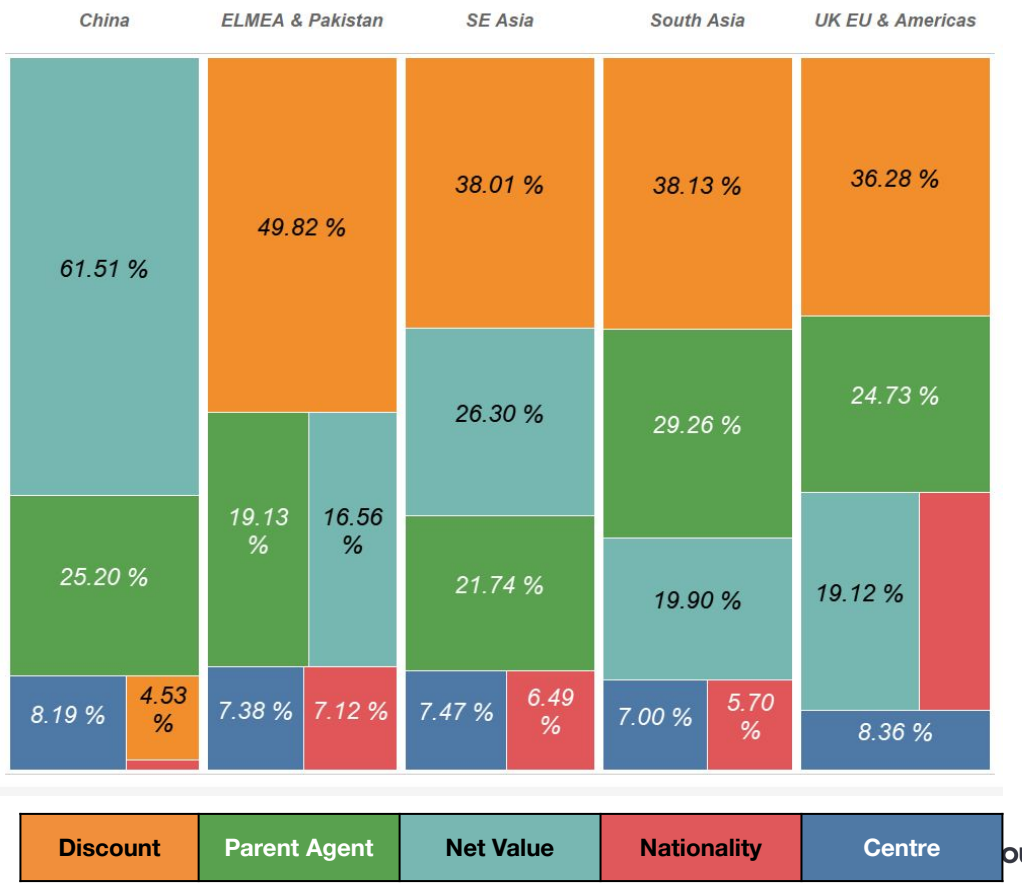
Feature Importance

Analysis: Feature importance, per area

Parent Agent (Group URN) added to the model

Agent variations contribute 19-29% to predictions, with South Asia most affected.

Discount has the highest impact in Pakistan, while **Net Value** dominates in China



Impact of Discount Across Regions

3 Analysis:

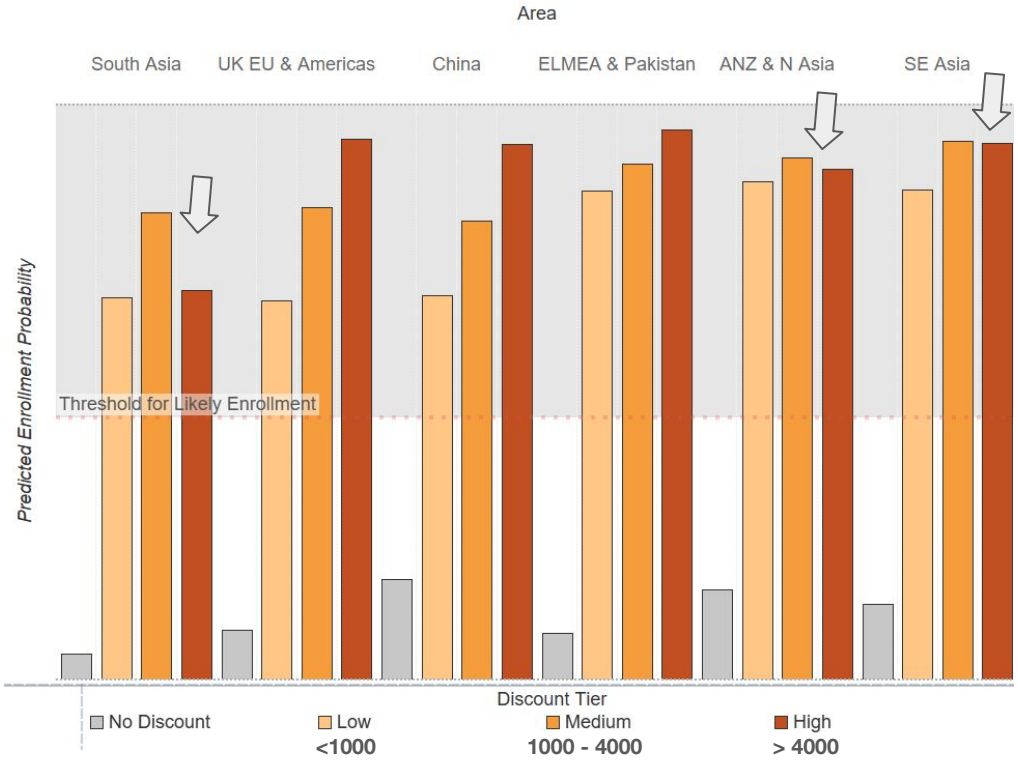
Evaluated **discount sensitivity per Area**

Diminishing marginal returns observed across regions.

South Asia and ANZ & Asia plateau, suggesting limited impact of further discounts.

UK, EU & Americas, China, and ELMEA show better responses to higher discount levels

Low-tier discounts still yield high enrollment probabilities, highlighting potential for **more efficient allocation** when combined with socio-economic insights.



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Agent Performance (Parent Agent Level Analysis)

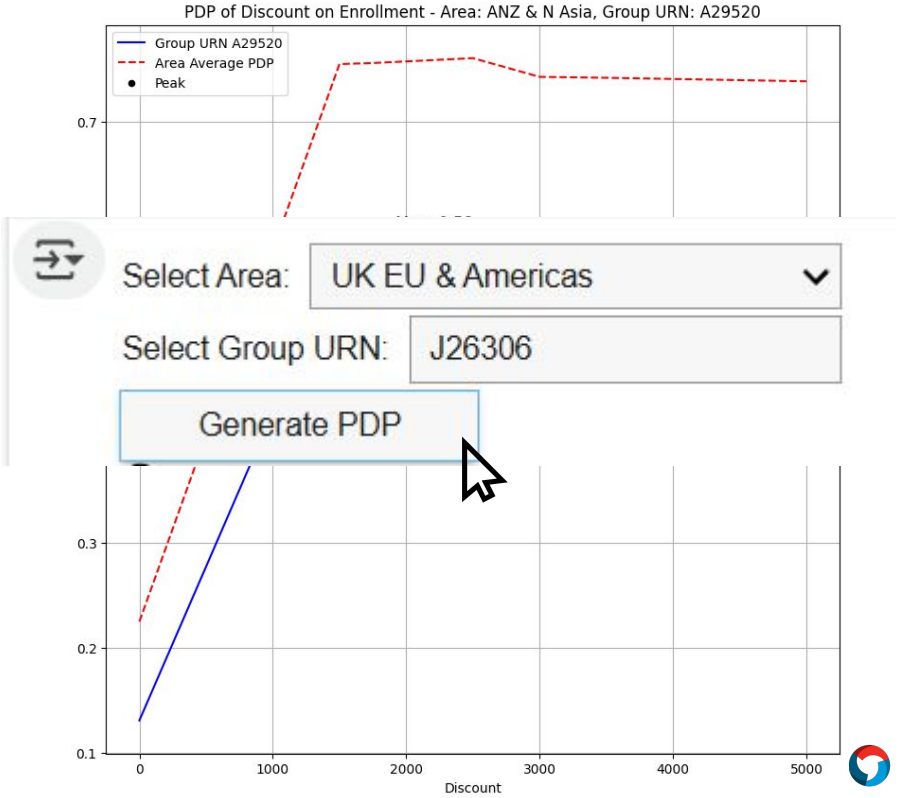
3 **Analysis:**

Available in the Python Notebook

Applies the enrollment model at the Parent Agent level (Group URN).

Enables direct comparison with the Area Average and peak enrollment probability.

Supports more granular insights into agent performance



Optimizing Agent Performance: Identifying Best Practices & Improvement Areas

ANZ & N Asia		UK EU & Americas		South Asia		China		ELMEA & Pakistan		SE Asia	
Top 5	Bottom 5	Top 5	Bottom 5	Top 5	Bottom 5	Top 5	Bottom 5	Top 5	Bottom 5	Top 5	Bottom 5
T25102	I12369	L22598	O24514	S30727	F20681	J24756	D27221	G25183	B25776	U13613	C10024
V11789	C1067	C31591	Y31287	I21639	V4750	S26815	H26002	C23327	O25811	V28168	G29261
J7734	A12814	A31743	I22194	E12609	M29111	S24396	H15431	U20664	C23318	F6995	D27017
B29849	S14337	G8315	I23083	A22473	B29757	H29288	S29589	E10347	C23329	C31617	E24124
F5767	R28083	S26462	E24037	C31344	P31831	S28280	S30879	E18853	G14680	E24324	Y26738

- **Best Practices**: Collect information of common traits / approaches among top performers to refine discount & recruitment strategies.
- **Challenges:** Collect information about challenges faced by bottom performers to assess potential interventions.

Current discount allocation

A total of £5,631,216 was 'spent' on 2126 students who confirmed their places, and £2,783,650 of this investment produced the 'return' of the student registering with the University (962 students). This represents a 49% full benefits realisation.

Despite 192 applicants passing the pathway course for Royal Holloway, 0 went on to register with the University. £644,285 of discount was allocated to students who confirmed places for Royal Holloway.

Despite 81 applicants passing the pathway course for Kingston University, only 1 went on to register with the University. £121,500 of discount was allocated to students who confirmed places on the pathway course for Kingston University.

100% of applicants to Lancaster University cancelled before arrival, however numbers are relatively low with only 260 offers made and £5,750 of discount allocated.

Insight: Study group could consider reducing the number of discounts offered to centres with a low conversion rate between confirmation and registration

Centre Discount Effectiveness

ROYAL HOLLOWAY UNIVERSITY OF LONDON

- 4th highest discount offered
- 192 Apps Passes
- 0 Registration
- £644,285 wasted

Kingston University London

- 81 Students Passed
- 1 Registration
- £121,500 wasted

Lancaster University

- 100% cancelled
- 260 offers made
- £5,750 wasted

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Logistic Regression Model for Student Registration

The goal of this model is predict if a student will register. Data preparation includes cleaning/filtering, resampling and splitting

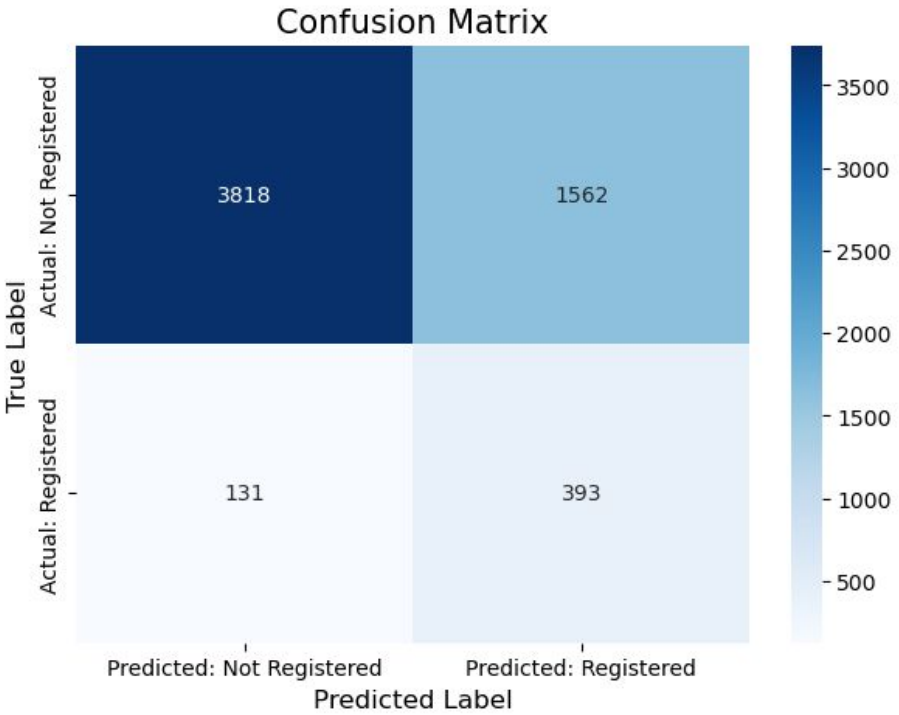
Model focus:

- Unique student IDs with offers.
- Key features (e.g. nationality, centre, etc)
- Predicting registration.
- Grouped smaller

Initial results:

- The matrix on the left shows that this model is 71% accurate.
- Correctly predicts that **3,818 students won't register** and **393 that will.**

This model to help us decide on what Students, StudyGroup, should be targeting.



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Logistic Regression Model Accuracy (ROC AUC)

All Features Model:

The ROC AUC score is approximately 0.82. This indicates a strong capability to separate registered and non-registered students. A value close to 1 represents excellent discrimination.

Important Features Model:

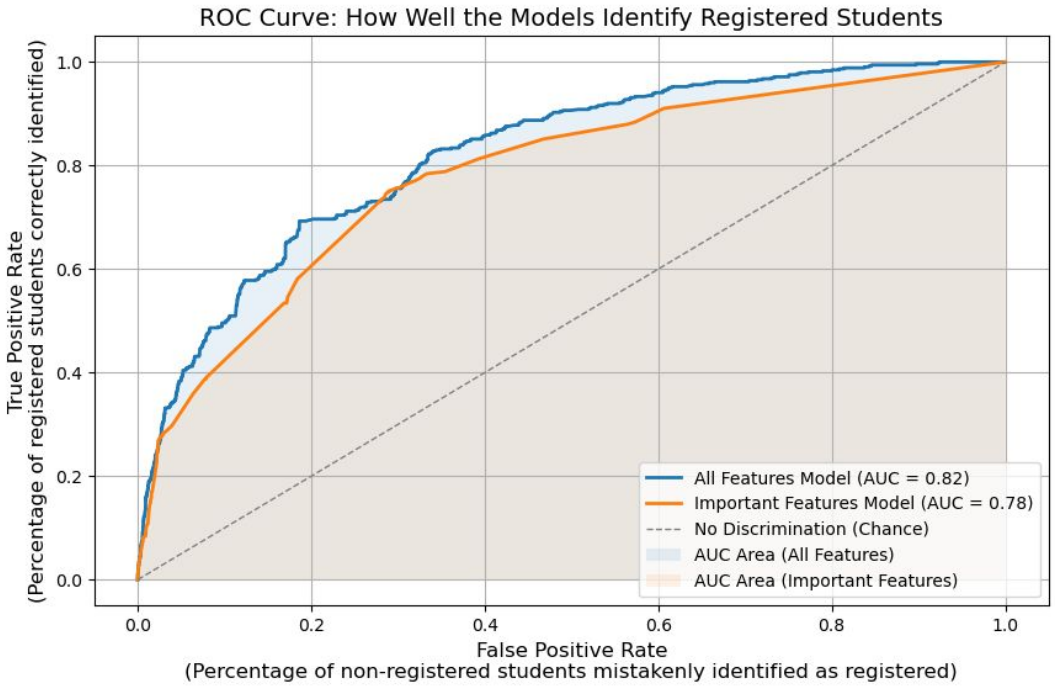
The ROC AUC score is approximately 0.78. Although slightly lower than the full model, this score still shows a reasonable performance with only the selected important features used for prediction.

Model Performance:

Effectively distinguishes between students who register and those who do not.

Interpretation:

A higher score indicates better performance in separating the two groups.



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Key Factors Driving Student Registration

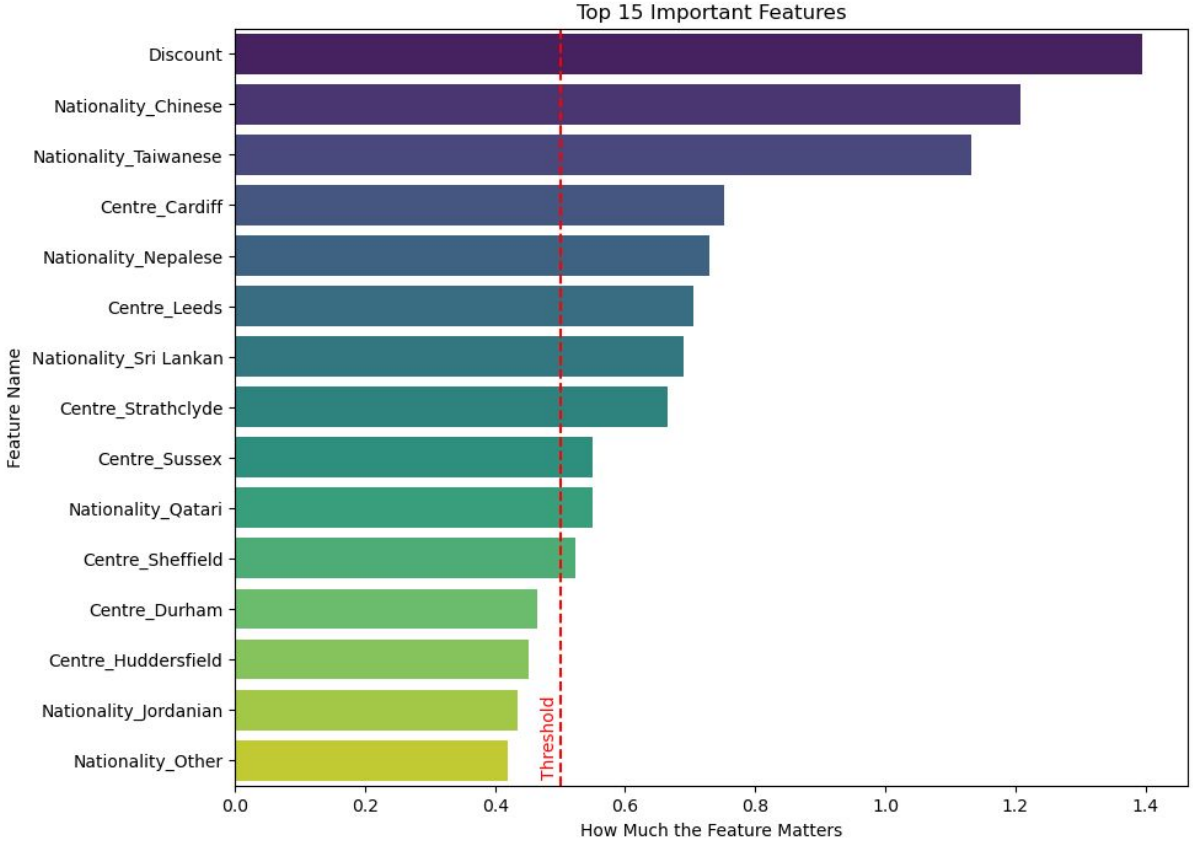
Discount is the most important feature in predicting registration, with the highest importance score of 1.40

Looking at nationality influence, Several nationalities have high importance scores, including:

- Chinese (1.20)
- Taiwanese (1.13)

Some study centres show higher importance in predicting registration:

- Cardiff (0.75)
- Leeds (0.70)



Summary of recommendations

Maximise volume of enrollments with Study Group:

- Focus discounts more on Foundation & Diploma courses
- Cap discounts to highly responsive areas (ANZ, Asia and SE Asia) at £1500 to avoid diminishing returns whilst maximising enrollments.
- For applicants from China, focus on Net Tuition Reduction and customising marketing materials to affordability

Maximise onward registrations with centres, ensuring maximum financial benefit to Study Group:

- Consider incentivising APD bookings where students typically require fewer discounts and have a higher conversion rate from offer to registration.
- Review discount allocation to lower performing centres (Royal Holloway, Kingston & Lancaster) and consider re-allocating to centres Cardiff, Leeds and Strathclyde where discounts have a demonstrated impact on registration.

Go further:

- Consider collecting additional student data, such as socioeconomic background and cancellation reasons to gain deeper insights into the factors influencing student decision-making.
- Collect more (qualitative) information on best practices and challenges of agents (by area) to find approaches to replicate / common limiting factors

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Thank you for listening.

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